

chap_pipps_comparaison

2025-07-09

Comparaison des stratégies (SI, HT) (PO, HT) et (SY, HT) par étude de Monte-Carlo

```
library(survey)

## Loading required package: grid
## Loading required package: Matrix
## Loading required package: survival
##
## Attaching package: 'survey'
## The following object is masked from 'package:graphics':
##
##     dotchart
library(sampling)

##
## Attaching package: 'sampling'
## The following objects are masked from 'package:survival':
##
##     cluster, strata
data(rec99); attach(rec99)

N <- nrow(rec99)
n <- 70
y <- LOGVAC
pik <- inclusionprobabilities(LOG,n)
pik.egales <- rep(n/N,N)
n.sim <- 10000
r1 <- replicate(n.sim, srswor(n,N))
ht1 <- apply(r1, 2, FUN=function(s,y,pik)
HTestimator(y[s==1],pik[s==1]),y,pik.egales)
r2 <- replicate(n.sim, UPpoisson(pik))
ht2 <- apply(r2, 2, FUN=function(s,y,pik)
HTestimator(y[s==1],pik[s==1]),y,pik)
r3 <- replicate(n.sim, UPsystematic(pik))
ht3 <- apply(r3, 2, FUN=function(s,y,pik)
HTestimator(y[s==1],pik[s==1]),y,pik)

# CV
sqrt(var(ht1))/mean(ht1)

## [1] 0.1896516
```

```
sqrt(var(ht2))/mean(ht2)
```

```
## [1] 0.1129701
```

```
sqrt(var(ht3))/mean(ht3)
```

```
## [1] 0.0776902
```

Pour aller plus loin avec R

```
set.seed(12345)
n.sim <- 10000
Prob <- matrix(0,nrow=n.sim,ncol=N)
for(i in 1:n.sim)
{
  s <- sample(1:N,n,prob=pik)

  Prob[i,1:N] <- (1:N %in% s)
}
pik_estimées <- apply(Prob,2,mean)
max(pik_estimées/pik)
```

```
## [1] 1.451524
```

```
#[1] 1.451524
```